# Smarter Shopping: Does Large Really Me..

Food shopping is always a mystery. The goal is to get the best for your buck but with figures and measurements and sizes ranging from small to large, how can you truly be sure. It can get a bit stirry. Usually, you associate visual perception to quantity. Large must be more and conversly, small must mean too little or not enough.

What about at a fish market or in the fish aisle? Rows of fish lined up on a bed of ice, you can't really pull out a measuring tape can you. Your best bet is to make a guess using the width, but is width and length a true indicator of mass?

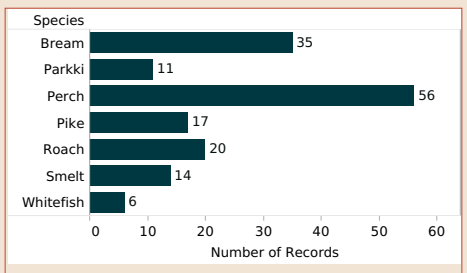
**Sample Size**

The following is an analysis of a sample dataset of 7 common fish species caught and measured fom the **Laengelmavesi Lake** in Tampere, Finland.

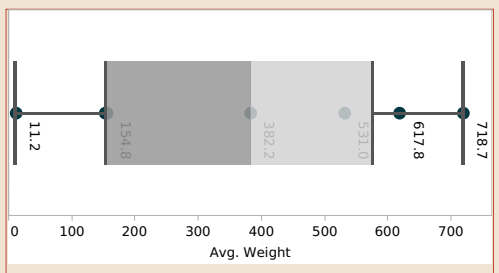
| Species | Avg. Length 1 (cm) | Avg. Length 2 (cm) | Avg. Length 3 | Avg. Width (cm) | Width Std. Dev | Avg. Weight (g) | Weight std. dev | Median Weight | Range of Weight | Min. Weight | Max. Weight |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Bream | 30.31 | 33.11 | 38.35 | 5.43 | 0.72 | 617.83 | 209.21 | 610 | 758 | 242 | 1,000 |
| Parkki | 18.73 | 20.35 | 22.79 | 3.22 | 0.64 | 154.82 | 78.76 | 145 | 245 | 55 | 300 |
| Perch | 25.74 | 27.89 | 29.57 | 4.75 | 1.77 | 382.24 | 347.62 | 207.5 | 1,094.1 | 5.9 | 1,100 |
| Pike | 42.48 | 45.48 | 48.72 | 5.09 | 1.14 | 718.71 | 494.14 | 510 | 1,450 | 200 | 1,650 |
| Roach | 20.65 | 22.28 | 24.97 | 3.66 | 0.69 | 152.05 | 88.83 | 147.5 | 390 | 0 | 390 |
| Smelt | 11.26 | 11.92 | 13.04 | 1.34 | 0.29 | 11.18 | 4.13 | 9.85 | 13.2 | 6.7 | 19.9 |
| Whitefish | 28.80 | 31.32 | 34.32 | 5.47 | 1.19 | 531.00 | 309.60 | 423 | 730 | 270 | 1,000 |

## Descriptive Statistics

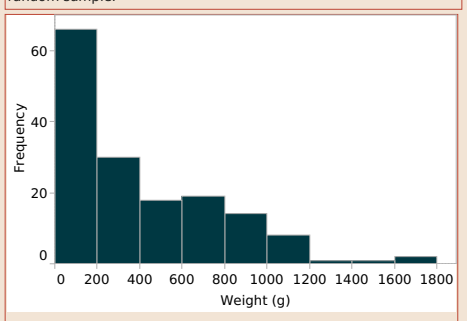
**Filter By Fish Species:**
All



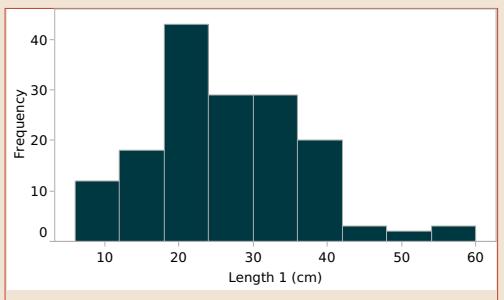

Number of observations per species in Laengelmavesi Lake in random sample.




Average of Weight. Details are shown for Species. The view is filtered on Species, which keeps 7 of 7 members.



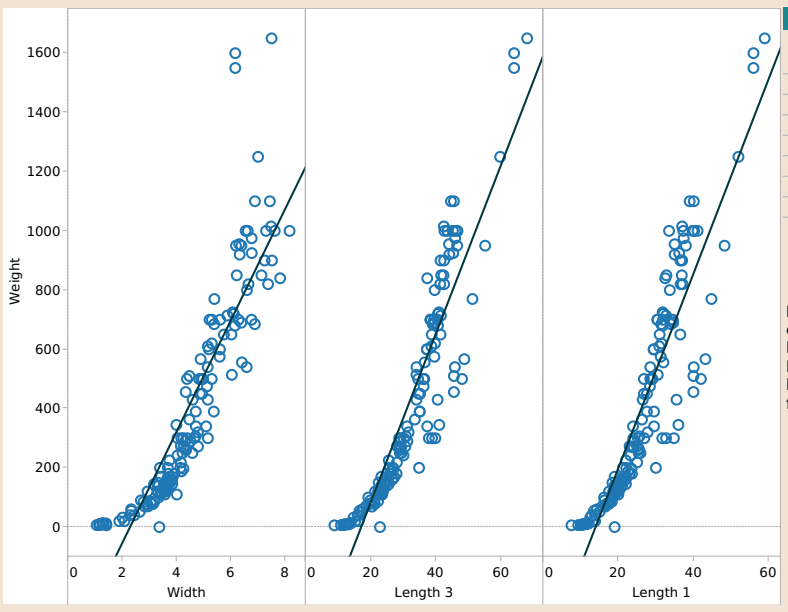

Histogram for Laengelmavesi Lake fish: Weights in gram (G).
Class size = 200, Skewed left




Histogram for Laengelmavesi Lake fish: Length 1 (cm) the length from nose to beginning of the tail in centimeters.
Class size = 200, Skewed right

## Inferential Statistics

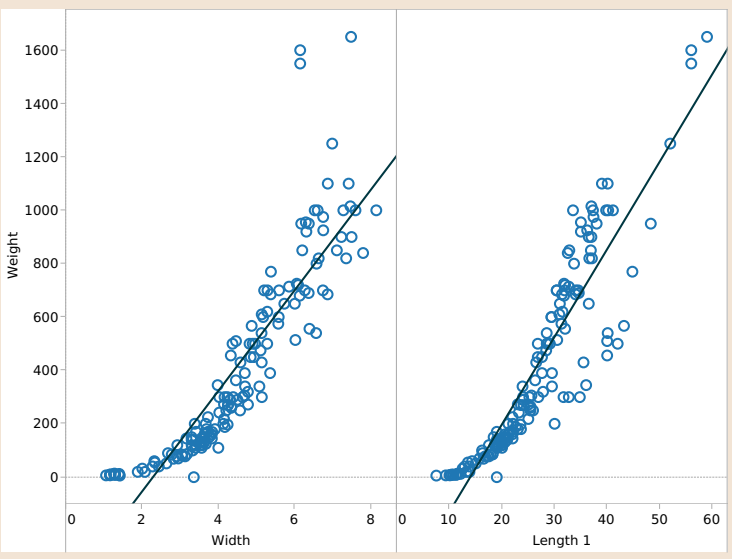
**Filter By Fish Species:**
All




**Correlation Coefficient Values, R**

| Species | R for Weight/W | R for Weight/L3 | R for Weight/L1 |
|---|---|---|---|
| Bream | 0.93 | 0.95 | 0.94 |
| Parkki | 0.99 | 0.97 | 0.97 |
| Perch | 0.96 | 0.96 | 0.96 |
| Pike | 0.92 | 0.98 | 0.97 |
| Roach | 0.92 | 0.92 | 0.91 |
| Smelt | 0.92 | 0.95 | 0.95 |
| Whitefish | 0.87 | 1.00 | 1.00 |

**Interpretation:** The general trend shows exponential increase in weight as either width, length 1 or length 3 increases. That is, where length 1 = the length from the nose to the beginning of the tail and length 3 = the length from the nose to the end of the tail.




### Does Large Really Mean More?

The average size of a market bream fish is 42.5 cm (their maximum recorded size is 90 cm). Without a tape measure and on a busy day, would a visual guess in selecting the largest fish help you get the most for your dollar?

Using the simple regression model we can assume that yes, width and length can be used to predict the weight of fish in the sample.

Weight vs. Maximal width as % of Length3. The data is filtered on Species, which keeps 7 of 7 members.

Linear Model: (1) Weight = 188.25 (width) + (-433.26), R-squared = 0.79 p-value = <0.0001
(2) Weight = 32.79 (length 1) - 462.38, R-squared = 0.84 p-value = <0.0001